

Proxy-Based Urban Change Prediction under Spatial Generalization: Improving Tail Reliability with Lightweight Calibration

Abstract

Predicting urban change from gridded Earth-observation proxies is increasingly important for urban monitoring, infrastructure assessment, and spatial decision support, but the hardest cases are not average fluctuations. They are rare local shocks, which make remotely sensed proxy targets such as Δ WorldPop and Δ VIIRS strongly long-tailed. Existing results can therefore be misleading for two reasons: row-wise splitting often overstates performance through spatial leakage, and average metrics understate failure in the tails, where monitoring risk is concentrated. Using CivitasGrid-TLP (2015–2024), we study long-tail spatial regression under leakage-aware GroupSplit(`grid_id`) and benchmark multiple tabular regressors together with two tail-oriented strategies: routing-based Mixture-of-Experts (MoE) and a lightweight post-hoc affine calibrator trained with an RMSE-anchored contrastive objective. The findings are clear. Δ WorldPop is consistently more predictable than Δ VIIRS, whose year-to-year differencing amplifies observational noise and leaves weak signal under realistic spatial generalization. Tail errors dominate the practical difficulty of the task. The anchored calibrator consistently improves tail reliability with minimal trade-off in overall fit across diverse baselines, whereas a two-expert MoE becomes less stable and underperforms under spatial generalization. These results highlight a practical lesson for applied remote sensing: when urban proxy mapping is transferred to unseen grids, lightweight calibration can improve reliability more effectively than adding routing complexity.

Keywords: applied remote sensing, urban proxy dynamics, nighttime lights, WorldPop, long-tail regression, spatial generalization

1. Introduction

1.1. Remote Sensing of Urban Dynamics: Why This Matters

Applied remote sensing is increasingly used to monitor urban change where official socioeconomic statistics are delayed, spatially coarse, or inconsistent across jurisdictions. Gridded Earth-observation products can provide spatially explicit signals of settlement intensity, population redistribution, and human activity at resolutions that are useful for urban planning and infrastructure assessment. In this setting, however, the most consequential errors are rarely concentrated in the average case. They occur when models miss abrupt local changes such as rapid expansion, stagnation, or decline in a small number of grid cells. For operational urban monitoring, reliability on these high-impact cells is therefore more important than small gains on already easy-to-predict regions.

1.2. The Challenge of Long-tail Remote-Sensing Proxy Dynamics

In this study, we examine year-to-year prediction of urban proxy dynamics at the grid level using Δ WorldPop and Δ VIIRS as targets. Both variables are derived from widely used remote-sensing products and are strongly long-tailed: most grids remain near zero from one year to the next, while a minority experience disproportionately large shifts. This distribution is not a minor nuisance. It defines the deployment setting. Rare shocks dominate both practical monitoring risk and model error, yet standard regression pipelines are typically tuned to improve average fit rather than tail reliability.

This setting exposes two gaps in current practice. First, reported performance can be overly optimistic when evaluation allows the same spatial units to appear in both training and test sets. In grid-based panel data, row-level random splitting can leak location-specific regularities across folds, inflating apparent generalization. For remote-sensing applications, this means a model may appear strong simply because it has effectively seen the same spatial footprint before. Second, even when overall metrics look acceptable, they can conceal systematic failure in the tails, where prediction quality matters most for identifying unusual urban change. A realistic study of urban proxy dynamics therefore needs both leakage-aware spatial evaluation and metrics that explicitly examine tail behavior.

Large-scale proxy data make this problem tractable. Official statistics are often delayed, spatially coarse, or inconsistent across regions, whereas

WorldPop and VIIRS provide scalable signals for fine-grained urban monitoring from remotely sensed observations. At the same time, differencing these proxies introduces additional observational noise, especially for VIIRS nighttime lights, making realistic benchmarking essential rather than optional.

1.3. Research Questions and Overview

Our goal is not to claim a fundamentally new predictive model class, but to resolve a practical mismatch between reported performance and real-world reliability in remote-sensing-enabled urban monitoring. Using CivitasGrid-TLP, we benchmark strong tabular regressors for long-tail spatial regression under GroupSplit(grid_id), a leakage-aware protocol that tests spatial generalization to unseen grids. This benchmark is paired with tail-focused evaluation so that average accuracy and rare-event reliability can be assessed together rather than conflated.

Within this framework, we study two complementary questions. The first is whether increasing model complexity through a two-expert Mixture-of-Experts improves long-tail prediction under realistic spatial splitting. The second is whether a lightweight post-hoc calibration layer can better align predictions with tail behavior without materially degrading overall fit. To answer the latter, we introduce an RMSE-anchored contrastive affine calibrator that is model-agnostic, plug-and-play, and applicable to strong non-differentiable tabular regressors such as tree ensembles. In remote-sensing terms, the question is whether reliability on rare urban change events is better improved through more complex routing or through better calibration of already informative predictors.

A central question of this paper is whether tail robustness in spatial long-tail regression is better addressed by routing-based decomposition or by lightweight post-hoc calibration under leakage-aware spatial evaluation. Our results favor the latter: the naive MoE does not rescue tail performance and introduces additional instability, whereas the anchored calibrator delivers more reliable tail behavior with minimal cost to overall accuracy. This makes the proposed framing especially relevant to applied remote sensing, where robust transfer to new geographic locations is often more valuable than marginal gains under permissive evaluation.

This work does not aim to propose a fundamentally new predictive model class. Instead, it focuses on identifying a practical mismatch between evaluation protocols and deployment conditions in spatial proxy prediction, and

provides a lightweight, model-agnostic correction strategy supported by systematic empirical evidence. We position this work as: (1) a leakage-aware evaluation benchmark, (2) a tail-focused empirical study, and (3) a practical calibration approach.

1.4. Our Contributions

- **Leakage-aware benchmark for long-tail urban dynamics:** We formulate grid-level prediction of $\Delta\text{WorldPop}$ and ΔVIIRS as a long-tail spatial regression problem for remote-sensing proxy monitoring and evaluate it under `GroupSplit(grid_id)`, reducing optimistic bias from spatial leakage and better matching deployment to unseen grids.
- **Systematic tail-focused evaluation:** We benchmark multiple strong tabular regressors using both overall and tail-specific metrics, showing that realistic spatial generalization is substantially harder than random-split performance suggests and that ΔVIIRS is markedly less predictable than $\Delta\text{WorldPop}$ under year-to-year remote-sensing differencing.
- **Lightweight model-agnostic calibration:** We propose an RMSE-anchored contrastive affine calibrator that can be attached post hoc to diverse regressors, including non-differentiable tabular models. The method consistently improves tail reliability with only minimal trade-off in overall metrics, which is attractive for operational remote-sensing workflows.
- **Actionable negative result on model complexity:** We show that a two-expert MoE does not improve this task under leakage-free evaluation, indicating that in long-tail spatial settings, added routing complexity does not automatically translate into better reliability for urban Earth-observation products.

Next, we review prior work on long-tail regression, mixture-of-experts learning under heterogeneity, and remote-sensing-based proxy indicators, which together motivate our benchmark setting.

2. Literature Review

2.1. Long-tail Regression

Traditional MSE/MAE-based regression models typically emphasize head (majority) samples at the expense of tail samples, resulting in reduced predic-

tive accuracy on tail samples, particularly for heavy-tailed data. To address
 this bias, Ren et al. proposed a *Contrastive Balanced MSE* approach that
 improves model performance on long-tailed and imbalanced samples [1]. This
 method not only optimizes the distance between $\hat{y}_i$ and y_i , but also max-
 imizes the distance between $\hat{y}_i$ and other samples' y_j , effectively reducing
 the dominant influence of head samples during training. Similarly, Yang
 et al. proposed *Deep Imbalanced Regression* [2], which combines *Label Dis-*
tribution Smoothing (LDS) and *Feature Distribution Smoothing* (FDS) to
 improve generalization under heavy-tailed targets. These complementary
 approaches motivate our setting, where long-tailed proxy dynamics interact
 with heterogeneous regional mechanisms and routing-based models.

2.2. Mixture-of-Experts

Yuksel et al. reviewed two decades of Mixture-of-Experts (MoE) research
 and applications [3]. Their work suggests that when data involve multiple
 regimes with heterogeneous mechanisms, a single model may be insufficient,
 but MoE with a soft routing gating network can substantially improve model
 fit by assigning each input to one or multiple experts. MoE can significantly
 improve fitting ability under heterogeneity; however, residual spikes may
 emerge when gating decisions are unstable.

While Yuksel et al. provide an overarching view of MoE as a divide-
 and-conquer framework, subsequent work formalizes MoE as a conditional
 mixture model. In Gormley and Frühwirth-Schnatter's research, MoE is
 formulated as a conditional mixture model where both mixing weights (π_k)
 and expert distribution parameters depend on covariates [4]. MoE can model
 heterogeneity and subgroup structure effectively; however, the stability of
 MoEs depends critically on the smoothness and identifiability of the gating
 model.

Building on this statistical formulation, recent surveys highlight how
 MoE has been adapted to diverse learning tasks including regression under
 heterogeneous conditional distributions. Nguyen and Chamroukhi provide a
 comprehensive overview of MoE variants across regression, classification, and
 clustering [5], systematically reviewing hierarchical MoE, mixture of linear
 experts, time-series mixtures, and mixture-of-regression models with gating
 networks. Through this review, they emphasize that gating routing (soft vs.
 hard) and gating-model complexity influence stability and interpretability. In
 our work, we implement a hierarchical MoE approach that divides head and
 tail into two separate regimes, each handled by a mixture of two experts.

142 Beyond point prediction, recent work explores distributional regression
 143 using MoE-based approaches, which is particularly relevant for heavy-tailed
 144 targets. Rügamer et al. propose an innovative approach for processing long-
 145 tailed and heavy-tailed data with outliers and shocks [6]. Rather than
 146 predicting $\mathbb{E}[Y|X = x]$, distributional regression models predict the full
 147 conditional distribution $Y|X = x$, which significantly enhances robustness
 148 and reliability. However, in practice, long-tail regression benchmarks often
 149 use target transformations and gating-based experts, which can suffer from
 150 boundary-induced residual spikes due to uncertain routing and transformation
 151 inversion.

152 Overall, existing MoE literature establishes the framework as a powerful
 153 strategy for learning heterogeneous regimes, and recent distributional exten-
 154 sions highlight its potential for robustness under heavy-tailed targets. However,
 155 most studies focus on modeling flexibility, inference, and application-level
 156 performance, leaving a gap in systematically characterizing and mitigating
 157 stability issues in long-tail regression. Specifically, when gating decisions
 158 are uncertain near boundaries and expert predictions are limited under dif-
 159 ferent target transformations, small differences can be amplified, leading to
 160 boundary-induced residual spikes and unacceptable worst-case errors. Moti-
 161 vated by this gap, our work empirically investigates stable MoE strategies
 162 under heavy-tailed proxy prediction and proposes an evaluation benchmark.

163 *2.3. Proxy-based Socioeconomic Indicators*

164 Beyond modeling strategies, our task depends on scalable, high-resolution
 165 proxies of economic dynamics, since official statistics are often unavailable at
 166 the grid level.

167 For population as an economic indicator, Paganelli notes that Adam
 168 Smith recognized population growth as a proxy for economic growth [7]: if a
 169 region can support population growth, its productivity is likely also increasing.
 170 Population change reflects aggregate outcomes of resource availability, labor
 171 demand, and wage dynamics rather than noise, providing meaningful signals
 172 for interpreting economic booms and recessions. This historical perspective
 173 validates our use of population-density data (WorldPop) when high-resolution
 174 official GDP data are unavailable. Peterson further emphasizes that popula-
 175 tion growth impacts vary by development level: in low-income economies it
 176 may dilute capital, while in high-income economies it can enable specializa-
 177 tion [8]. This heterogeneity motivates training specialized experts for different
 178 development contexts.

Regarding nighttime luminosity as an economic proxy, Chen and Nordhaus demonstrate that nighttime light intensity is an objective, globally-available, spatially high-resolution indicator that can be used to replace or supplement traditional economic statistics [9]. Their systematic analysis of GDP and luminosity across 12,393 globally distributed $1^\circ \times 1^\circ$ grid cells (1992–2008) found a notably strong correlation ($R^2 = 0.97$) between illuminated area and GDP, validating VIIRS Nighttime Light Intensity as a proxy for economic dynamics. However, at finer spatial scales, the relationship becomes noisier and less reliable, especially in low-output regions. Measurement error in log-luminosity across satellites introduces substantial uncertainty (SE 0.28–0.59), corresponding to $\geq 25\%$ proxy error. Thus, a more robust long-tail prediction model is essential.

Prior work primarily uses population or luminosity to approximate static economic output levels, but urban planning often requires predicting changes rather than absolute levels. This motivates defining targets as year-to-year differences rather than absolute values.

2.4. Literature Review Summary

In summary, prior work validates the use of population and nighttime luminosity as scalable remote-sensing proxies of urban and economic activity and extensively explores MoE for learning under heterogeneity. Long-tail regression research shows that standard MSE/MAE training biases models toward head samples, and remedies such as Balanced MSE and distribution smoothing (LDS/FDS) can improve tail performance under heavy-tailed targets. However, existing studies rarely provide unified benchmarks for grid-level proxy dynamics prediction that simultaneously reflect remote-sensing observation noise, transportation morphology features, long-tailed targets, and leakage-aware evaluation. To address this gap, we introduce CivitasGrid-TLP as a benchmark dataset and investigate robust long-tail modeling strategies with controlled MoE and contrastive learning frameworks.

3. Dataset: CivitasGrid-TLP

CivitasGrid-TLP is a grid-level panel dataset designed for proxy-based urban dynamics prediction from heterogeneous geospatial observations [10]. It integrates transportation network features and remotely sensed proxy signals (WorldPop population density and VIIRS nighttime light) across thousands

of city grids (30"–60" resolution), supporting an applied remote-sensing view of urban change monitoring rather than a purely tabular benchmark.

3.1. Data Sources and Grid Definition

We define the basic spatial unit as a grid cell (30"–60"), and constructed a panel dataset spanning 2015–2024. Each grid-year record links a spatial cell with a calendar year, enabling year-to-year dynamics prediction.

Source	Type	Resolution	License
OpenStreetMap	Transportation	N/A	ODbL
VIIRS V2	Nighttime Light	15"	CC0
WorldPop	Population Density	1 km	CC BY

Table 1: Data sources and specifications

As shown in Table 1, we extracted transportation features from OpenStreetMap and linked them with proxy signals including VIIRS nighttime light intensity and WorldPop population density. Unlike existing work, which typically focuses on coarse administrative regions, CivitasGrid-TLP provides a grid-based benchmark at kilometer-level resolution, supporting remote-sensing-oriented evaluation across regions and countries.

3.2. Targets and Feature Groups

Model inputs in CivitasGrid-TLP are organized into three groups: (i) transportation network features, (ii) lagged proxy levels, and (iii) categorical descriptors. Transportation features are derived from OpenStreetMap (OSM) and aggregated into seven morphology categories capturing road density, hierarchy, and intersection structure. To ensure comparability across grids of different spatial resolutions, all OSM-derived features are normalized by grid area (per km²). When OSM coverage is absent or insufficient, transportation features are set to zero, indicating unmapped road infrastructure.

For proxy signals, we include previous-year absolute levels of WorldPop population density and VIIRS nighttime light intensity as minimal temporal context. The prediction targets are year-to-year changes: $\Delta\text{WorldPop}$ and ΔVIIRS , which emphasize local dynamics while reducing baseline-level heterogeneity across regions.

239 Finally, categorical descriptors including country code, city type, and func-
 240 tional zone characterize heterogeneous regional contexts and enable stratified
 241 evaluation across development regimes.

242 3.3. Data Definition and Calculation

243 We define the prediction task as grid-level, year-to-year dynamics forecast-
 244 ing of proxy economic signals. Given features and lagged proxy observations
 245 for a grid cell, the goal is to predict proxy signal changes from year $t - 1$ to
 246 year t .

247 Specifically, we forecast two targets: population density change ($\Delta\text{WorldPop}$)
 248 and nighttime light change (ΔVIIRS), defined as:

$$\Delta\text{WorldPop} = \text{WorldPop}_t - \text{WorldPop}_{t-1} \quad (1)$$

$$\Delta\text{VIIRS} = \text{VIIRS}_t - \text{VIIRS}_{t-1} \quad (2)$$

250 Differencing captures local urban dynamics and reduces baseline differences
 251 across regions. Most grid cells experience minor year-to-year fluctuations,
 252 while a non-negligible subset undergoes abrupt shocks. From a remote-sensing
 253 perspective, these differenced products emphasize change detection but also
 254 amplify sensor- and preprocessing-related noise, which motivates strategies
 255 prioritizing tail and worst-case performance.

256 3.4. Quality Control and Filtering Tiers

257 Given substantial variation in data quality and completeness across regions
 258 and time, we adopt a tiered filtering procedure to construct subsets with
 259 different quality-coverage trade-offs. Transportation features are grouped
 260 into seven feature categories. For each grid-year record, we count available
 261 (non-missing) OSM feature categories.

262 We define two filtering tiers based on transportation feature completeness.
 263 Records with fewer than three available OSM feature categories are excluded
 264 due to insufficient information. Remaining records form the Medium-Quality
 265 (MQ) dataset. A stricter High-Quality (HQ) dataset is further defined by
 266 retaining records with at least four available OSM feature categories. This
 267 tiered design enables systematic evaluation under different quality constraints
 268 while maintaining transparency and reproducibility. We begin with 7,500
 269 candidate grids. To focus on economically meaningful urban dynamics and
 270 exclude extremely sparse areas with limited analytical value, we first remove
 271 grids with population density below 5 km^{-2} . This leaves 6,178 unique grids in

272 CivitasGrid-TLP v1.1.0. We then apply the MQ/HQ filtering criteria based
 273 on OSM feature completeness. The table below reports the resulting grid
 counts after each stage.

Dataset Tier	Grid Count
Initial Candidates	7,500
Population Masked	6,178
Medium-Quality (MQ)	4,293
High-Quality (HQ)	2,833

Table 2: Dataset filtering results (grid counts)

274
 275 In the next subsection, we report the resulting sample sizes, geographic
 276 coverage, and long-tail diagnostics for the MQ and HQ subsets.

277 3.5. Dataset Statistics and Long-Tail Diagnostics

278 After removing candidate grids with population density below 5 km^{-2} ,
 279 CivitasGrid-TLP v1.1.0 contains 6,178 unique grid cells observed over 10 years
 280 (2015–2024), yielding 61,780 grid-year records in total. Applying the OSM-
 281 completeness filters further retains 42,930 records in the Medium-Quality (MQ)
 282 subset and 28,330 records in the High-Quality (HQ) subset, corresponding to
 283 4,293 and 2,833 unique grids, respectively.

284 The dataset covers 85 countries (HQ) or 100 countries (MQ) with diverse
 285 regional contexts, including nation code, region type, and city type as categor-
 286 ical descriptors. This diversity introduces heterogeneous feature distributions
 287 and motivates generalization evaluation across regions. Top 5 countries by
 288 grid count:

Rank	Nation Code	HQ Grid Count	MQ Grid Count
1	USA	1,715	2,523
2	CHN	401	581
3	IND	90	186
4	JPN	87	120
5	CAN	43	68
Total	N/A	2,833	4,293

Table 3: Geographic distribution: top-5 countries by grid count

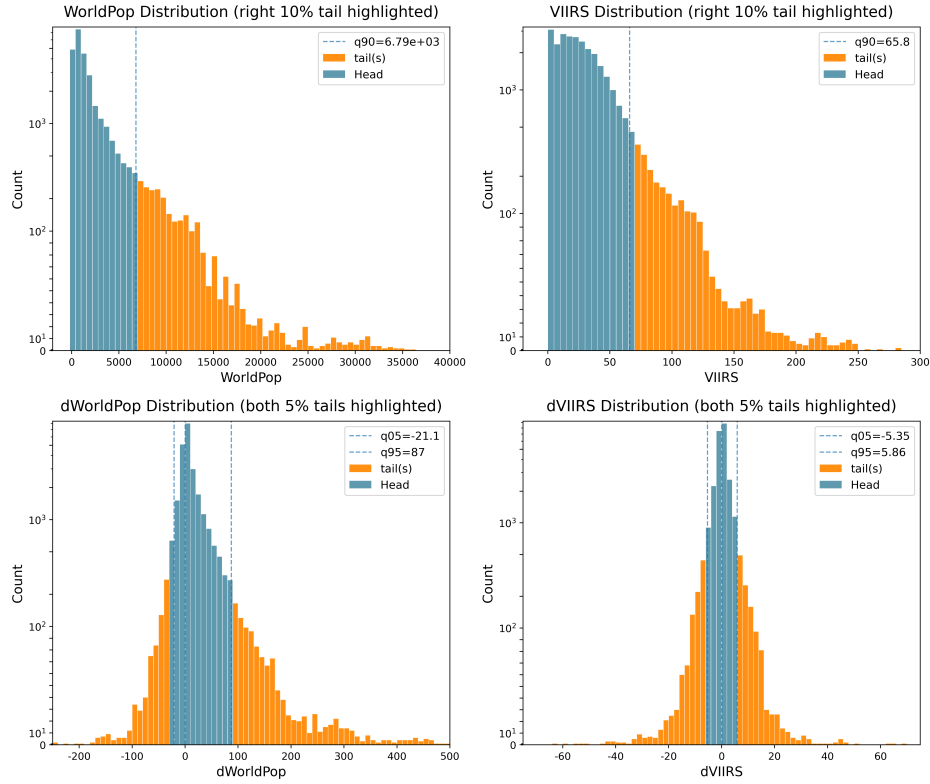


Figure 1: Distribution of proxy signals and their year-to-year changes, illustrating the long-tailed nature of both absolute levels and changes. The y-axis uses a piecewise log scale: values below 100 are plotted linearly, while values above 100 use a logarithmic scale to reveal tail frequencies.

289 As shown in Table 5, both $\Delta\text{WorldPop}$ and ΔVIIRS exhibit high variance
 290 and strong positive kurtosis, indicating heavy-tailed dynamics with frequent
 291 small fluctuations and occasional extreme shocks.

Target	Mean	Std. Dev.	Min	5%	25%	Median
$\Delta\text{WorldPop}$	16.37	52.54	-733.6	-21.10	-1.258	4.195
ΔVIIRS	0.190	4.333	-63.60	-5.354	-1.042	0.100

Table 4: Target summary statistics (part 1)

Target	75%	95%	Max	Skewness	Kurtosis
$\Delta\text{WorldPop}$	20.60	86.97	930.9	5.520	65.96
ΔVIIRS	1.402	5.864	68.86	0.174	29.41

Table 5: Target summary statistics

292 Figure 1 visualizes the distributions of proxy signals and their year-to-
 293 year changes for the HQ dataset. While absolute levels are right-skewed,
 294 differenced targets exhibit clear heavy-tailed patterns: most values concentrate
 295 near zero, while a substantial fraction lies far beyond. Such long-tailed targets
 296 pose challenges for average-error-optimized regression models and for remote-
 297 sensing change analysis more broadly, because rare but meaningful local
 298 anomalies are precisely the cases users care about.

299 **Tail and Head Split.** For each target y (ΔVIIRS and $\Delta\text{WorldPop}$), we
 300 define tail samples as grid-year records with absolute magnitude $|y|$ in the
 301 top 5% of the empirical distribution. This symmetric definition captures both
 302 positive and negative shocks, focusing on rare high-magnitude events. We
 303 use this tail subset throughout evaluations to characterize long-tail behavior
 304 and analyze model reliability under extreme events.

305 3.6. Spatial Coverage and Sampling Bias

306 Figure 2 shows the global spatial distribution of grids in CivitasGrid-TLP.
 307 Coverage is comprehensive but nonuniform: grids concentrate in regions with
 308 well-mapped road networks and stable proxy availability (North America,
 309 Europe, East and South Asia), while sparse in low-coverage areas. This imbal-
 310 ance is expected for datasets combining volunteered geographic information
 311 with remote sensing.

312 Non-uniform coverage introduces sampling through two channels. First,
 313 OSM completeness varies substantially across and within countries, affecting
 314 transportation feature availability. Second, quality-control filtering further

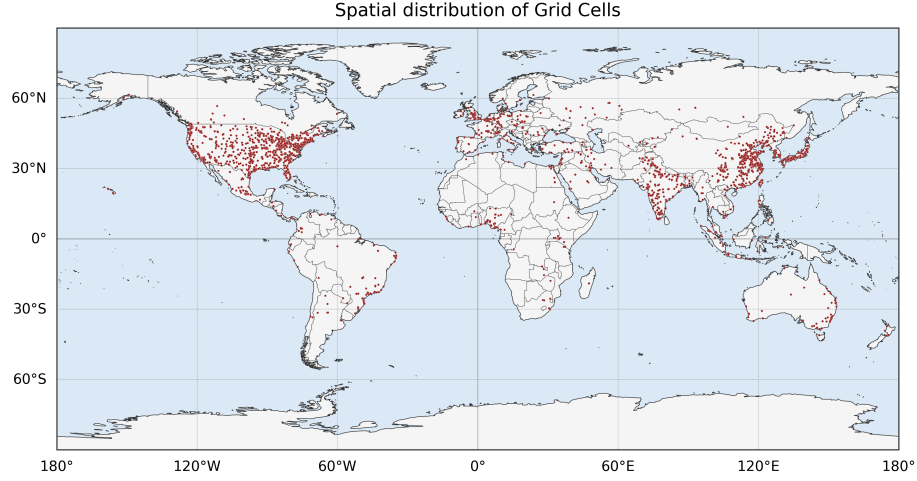


Figure 2: Global spatial distribution of grids in CivitasGrid-TLP, showing coverage concentration in well-mapped urbanized regions.

excludes records with insufficient features, implicitly favoring well-mapped urbanized regions and disadvantaging rural areas. HQ/MQ subsets therefore over-represent documented regions while under-representing rural contexts.

These long-tail and coverage challenges motivate our modeling framework and evaluation protocol.

4. Methodology

This section introduces the benchmark baselines and our controlled experimental variants, including an objective-level contrastive learning (CL) calibration layer and a two-expert MoE design for improving tail and worst-case performance. The methodological emphasis is practical: we evaluate whether common machine-learning tools can be made more reliable for remote-sensing proxy monitoring under realistic geographic transfer.

4.1. Problem Setup and Tail Definition

We evaluate two strategies for improving tail performance—Mixture-of-Experts and contrastive learning calibration—on the CivitasGrid-TLP dataset. We consider a grid-level regression problem where each sample has features x_i consisting of transportation morphology, lagged proxy levels, and categorical descriptors, while the target is the year-to-year change in the proxy signal.

333 The targets exhibit long-tailed distributions, with most samples near zero and
 334 a small but significant fraction at extreme values. In remote-sensing terms,
 335 these tail cases correspond to unusually large local deviations in the observed
 336 proxy fields. We define tail samples as those in the lower 5% or upper 5% of
 337 the empirical distribution:

338 *Definition 1 (Tail Region)*.. Let q_{low} and q_{high} denote the 5th and 95th per-
 339 centiles of the training targets. We define the tail region as samples where
 340 $y < q_{\text{low}}$ or $y > q_{\text{high}}$.

341 *Definition 2 (Tail RMSE)*.. Tail RMSE is computed as the root mean squared
 342 error restricted to samples in the tail region, i.e.,

$$\text{RMSE}_{\text{tail}} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} (y_i - \hat{y}_i)^2}. \quad (3)$$

343 *Definition 3 (Spatial Leakage)*.. Spatial leakage occurs when spatial units
 344 (`grid_id`) appear in both training and test sets, allowing models to exploit
 345 location-specific correlations that do not generalize to unseen grids.

346 *Definition 4 (Anchored Contrastive Objective)*.. The calibrator is trained
 347 using a combination of RMSE loss and a contrastive loss normalized by the
 348 base model’s reference scale, ensuring balanced optimization between global
 349 fit and tail discrimination. In our implementation, this reference scale is
 350 computed from the base calibrator $(a, b) = (1, 0)$ on the training split.

$$\begin{aligned} i \in \mathcal{T}_{\text{VIIRS}} &\iff \begin{cases} \Delta \text{VIIRS}_i \leq -5.35, \\ \Delta \text{VIIRS}_i \geq 5.86, \end{cases} \\ i \in \mathcal{T}_{\text{WorldPop}} &\iff \begin{cases} \Delta \text{WorldPop}_i \leq -21.1, \\ \Delta \text{WorldPop}_i \geq 87. \end{cases} \end{aligned} \quad (4)$$

351 where the thresholds correspond to empirical percentiles. In CivitasGrid-
 352 TLP, each sample i is a *grid-year* record: a fixed spatial grid at year t , with
 353 features from transportation morphology, lagged proxy levels, and categorical
 354 descriptors, and targets as first differences $\Delta s_{i,t} = s_{i,t} - s_{i,t-1}$ for proxy signal
 355 s .

356 To ensure reproducibility and prevent leakage, all percentile thresholds
 357 are computed on the training split only and applied unchanged to validation
 358 and test splits.

359 We treat ΔVIIRS and $\Delta\text{WorldPop}$ as separate targets with independent
 360 tail sets. In evaluation, we report overall metrics on the full set and tail-
 361 specific metrics on the subset, since our goal is improving reliability under
 362 rare high-magnitude shocks.

363 4.2. Baseline Model

364 We choose XGBoost as our primary baseline because it performs well
 365 on tabular data through its ability to capture nonlinearities and feature
 366 interactions. The implementation is mature, efficient, and scalable, providing
 367 stability and interpretability.

368 XGBoost is a regularized gradient boosting framework that builds addi-
 369 tive decision tree ensembles by iteratively fitting new trees to negative loss
 370 gradients using second-order Hessian information.

371 We also include Random Forest as a non-boosting classical baseline that
 372 provides complementary performance. Additionally, we include several stan-
 373 dard baselines (Ridge Regression, Elastic Net, SVR, Extra Trees Regressor,
 374 and MLP) in controlled experiments to ensure our conclusions are not model-
 375 specific.

376 4.3. Contrastive Learning Evaluation

377 We evaluate *objective-level* contrastive learning (CL) as a generic tail-aware
 378 signal. Because many competitive tabular regressors are non-differentiable
 379 (e.g., random forests, SVR), we apply CL through a lightweight *post-hoc affine*
 380 *calibration* layer that can be trained on top of *any* baseline predictor.

381 Given a trained baseline $f(\cdot)$, we first obtain its predictions $\hat{y}^{(0)} = f(x)$.
 382 We then train an affine calibrator on the training split only:

$$\tilde{y} = a \hat{y}^{(0)} + b, \quad (5)$$

383 and evaluate $\tilde{y}$ on the held-out test split. We consider three configurations
 384 per baseline:

- 385 • **No-CL:** raw baseline prediction $\hat{y}^{(0)}$ (no calibrator);
- 386 • **CL-only:** calibrator trained using the contrastive objective $\mathcal{L}_{\text{con}}$, de-
 387 noted as (Baseline)_CL;
- 388 • **RMSE+CL (mixed):** calibrator trained using a normalized mixture
 389 of RMSE and $\mathcal{L}_{\text{con}}$, denoted as (Baseline)_RMSE_CL.

390 This design enables a controlled comparison of objectives while keeping the
 391 underlying baseline model and data splits fixed.

392 4.3.1. Contrastive objective

393 Given a mini-batch $\mathcal{B}$, we treat the calibrated prediction $\tilde{y}_i$ as the anchor
 394 and its paired ground-truth y_i as the positive, while other y_j in the batch serve
 395 as in-batch negatives. Using an L1 distance $d(\tilde{y}, y) = |\tilde{y} - y|$ and temperature
 396 T , we define

$$p_i = \frac{\exp(-|\tilde{y}_i - y_i|/T)}{\sum_{j \in \mathcal{B}} \exp(-|\tilde{y}_i - y_j|/T)}, \quad (6)$$

397 and the contrastive loss

$$\mathcal{L}_{\text{con}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} -\log p_i. \quad (7)$$

398 In our implementation, the temperature is set adaptively from the training
 399 target scale:

$$T = \max(\text{Std}(y_{\text{train}}) \cdot \kappa, 10^{-3}), \quad \kappa = 1.0. \quad (8)$$

400 4.3.2. Normalized mixed objective (RMSE+CL)

401 To combine RMSE with the contrastive loss in a numerically stable way,
 402 we optimize a normalized mixture on the calibrator parameters (a, b) :

$$\mathcal{L}_{\text{mix}} = w_{\text{rmse}} \cdot \frac{\text{RMSE}}{\text{RMSE}_0} + w_{\text{cl}} \cdot \frac{\mathcal{L}_{\text{con}}}{\mathcal{L}_{\text{con},0}}, \quad (9)$$

403 where $(w_{\text{rmse}}, w_{\text{cl}}) = (0.05, 0.95)$. The normalization constants are computed
 404 on the training split using the *initial* calibrator $(a, b) = (1, 0)$: RMSE_0 is
 405 computed exactly, while $\mathcal{L}_{\text{con},0}$ is estimated by subsampling at most 3000
 406 training samples and averaging the batch-wise contrastive loss using a batch
 407 size of 512 to avoid an $O(N^2)$ full-dataset computation.

408 4.3.3. Optimization details

409 We optimize (a, b) with Adam (betas 0.9/0.999, $\epsilon = 10^{-8}$) for 40 epochs,
 410 batch size 128, and learning rate 0.05. This calibrator training is performed
 411 independently for each baseline and target using training-split predictions
 412 only, and then applied unchanged to validation/test predictions.

413 4.4. Two-expert MoE framework

414 As an architecture-level baseline to address heavy-tailed regression targets,
 415 we adopt a two-expert mixture-of-experts (MoE) design that blends a *body*

regressor with a *tail* regressor through a learned soft gate. For a sample (x_i, y_i) , the MoE prediction is

$$\hat{y}_i = (1 - g(x_i)) \hat{y}^{(b)}(x_i) + g(x_i) \hat{y}^{(t)}(x_i), \quad (10)$$

where $\hat{y}^{(b)}(\cdot)$ and $\hat{y}^{(t)}(\cdot)$ denote the body and tail experts, and $g(x) \in (0, 1)$ is the gating probability. We implement all three components with XGBoost models, and use the same train/validation/test splits and preprocessing pipeline to ensure a fair comparison.

4.4.1. Body Expert

The body expert is a standard XGBoost regressor trained to predict the original target y :

$$\min_{\theta_b} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} \ell(y_i, \hat{y}^{(b)}(x_i; \theta_b)) + \Omega(\theta_b), \quad (11)$$

where $\Omega(\cdot)$ is the regularization term used by XGBoost. In our implementation, the body expert is trained on the full training split (rather than a filtered subset), with early stopping on the validation split when enabled.

4.4.2. Tail Expert (signed-log target space)

To stabilize extreme target target magnitudes, the tail expert is trained in a signed-log space:

$$z = \text{signedlog1p}(y) = \text{sign}(y) \log(1 + |y|). \quad (12)$$

The tail expert is an XGBoost regressor trained to predict z on the full training split:

$$\min_{\theta_t} \sum_{(x_i, y_i) \in \mathcal{D}_{\text{train}}} \ell(z_i, \hat{z}^{(t)}(x_i; \theta_t)) + \Omega(\theta_t), \quad (13)$$

where $z_i = \text{signedlog1p}(y_i)$. At inference time, we map predictions back to the original scale via the inverse transform

$$\hat{y}^{(t)} = \text{signedexpm1}(\hat{z}) = \text{sign}(\hat{z}) (\exp(|\hat{z}|) - 1). \quad (14)$$

435 4.4.3. Gating model

436 The gating model predicts whether a sample belongs to the tail region
 437 defined by training-split quantiles (Sec. 4.1). Let $(q_{0.05}, q_{0.95})$ be the 5th/95th
 438 percentiles computed on the training split. We define the tail indicator

$$r_i = \mathbb{I}(y_i \leq q_{0.05}) \vee \mathbb{I}(y_i \geq q_{0.95}), \quad (15)$$

439 and train an XGBoost binary classifier $g(x)$ by minimizing binary cross-
 440 entropy:

$$\min_{\phi} \sum_{i \in \mathcal{D}_{\text{train}}} \left[-r_i \log g(x_i; \phi) - (1 - r_i) \log(1 - g(x_i; \phi)) \right]. \quad (16)$$

441 At inference time, $g(x)$ provides a soft weight that smoothly blends body-
 442 like and tail-like behaviors without hard routing.

443 4.5. Evaluation protocol

444 This section presents technical details of our evaluation pipeline and
 445 metrics reported.

446 4.5.1. Data Splitting

447 We evaluate under two splitting strategies to assess both i.i.d. performance
 448 and spatial generalization robustness.

- 449 1. **Random Split (70/15/15):** Randomly partition samples into 70%
 450 training, 15% validation, and 15% test.
- 451 2. **GroupSplit(grid_id):** Split by grid_id to prevent spatial leakage: all
 452 samples from a grid belong to exactly one of train/val/test. Test grids
 453 are held out, and remaining grids are split into train and validation.

454 For each target, samples with missing labels are removed after splitting
 455 (filtering occurs within each split, per-target). All preprocessing (imputa-
 456 tion, scaling, one-hot encoding) is fit on training data only and applied to
 457 validation/test.

458 4.5.2. Metric Reporting

459 We report standard regression metrics including MAE, RMSE, and R^2 . To
 460 summarize worst-case behavior, we additionally report high-quantile absolute
 461 error, namely the 95th and 99th percentiles of $|y - \hat{y}|$ on the evaluation set.
 462 For each method and each split, we compute metrics on the test set and
 463 report them with consistent prefixes for overall vs. tail evaluation (see below).

4.5.3. Tail Evaluation

Tail samples are defined as described in Sec. 4.1. In all experiments, the tail thresholds (e.g., quantiles) are computed using the training split only and then applied unchanged to validation/test to avoid leakage. We report the tail rate in the test split and compute all metrics on both: (i) the full test set (**All_**), and (ii) the tail subset (**Tail_**).

5. Results

In this section we present and analyze experimental results across two targets and two splitting strategies. The main question is not only which method scores best on average, but which one produces more reliable urban proxy maps when transferred to unseen grids. We find that contrastive learning (CL) calibration substantially improves tail performance, while the two-expert MoE approach underperforms the baseline.

5.1. Benchmark Performance (Baseline Models)

Table 6: Benchmark performance on Δ VIIRS (dVIIRS)

Split	Model	R^2	RMSE	R^2_{tail}	RMSE _{tail}
RandomSplit	XGBoost	0.026	4.388	0.141	11.414
	Ridge	0.008	4.427	0.017	12.215
	ElasticNet	-0.000	4.446	-0.002	12.332
	SVR	-0.047	4.549	-0.014	12.403
	Random Forest	0.066	4.296	0.144	11.395
	ExtraTrees	0.014	4.414	0.131	11.485
	MLP	0.010	4.424	0.020	12.193
GroupSplit	XGBoost	0.058	4.675	0.137	12.869
	Ridge	0.025	4.757	0.030	13.648
	ElasticNet	-0.000	4.817	-0.000	13.856
	SVR	-0.009	4.838	0.022	13.706
	Random Forest	0.105	4.558	0.173	12.604
	ExtraTrees	0.093	4.587	0.168	12.639
	MLP	0.038	4.725	0.053	13.486

Benchmark results for both Δ VIIRS and Δ WorldPop are shown in Tables 6 and 7. Several patterns emerge:

First, Δ VIIRS is consistently difficult to predict. Across both RandomSplit and GroupSplit, most models remain near zero or below zero in R^2 , indicating that year-to-year light changes carry little recoverable signal beyond a mean predictor in our current feature setting. A plausible explanation is that

Table 7: Benchmark performance on Δ WorldPop (dWorldPop)

Split	Model	R^2	RMSE	R^2_{tail}	RMSE _{tail}
RandomSplit	XGBoost	0.836	21.976	0.832	61.048
	Ridge	0.354	43.593	0.366	118.483
	ElasticNet	0.274	46.197	0.215	131.789
	SVR	0.440	40.597	0.373	117.766
	Random Forest	0.784	25.209	0.770	71.417
	ExtraTrees	0.843	21.499	0.826	62.033
	MLP	0.766	26.239	0.791	68.011
GroupSplit	XGBoost	0.527	38.462	0.563	108.125
	Ridge	0.288	47.194	0.208	145.595
	ElasticNet	0.211	49.658	0.052	159.212
	SVR	0.374	44.234	0.283	138.528
	Random Forest	0.511	39.116	0.573	106.848
	ExtraTrees	0.466	40.875	0.513	114.195
	MLP	0.416	42.739	0.654	96.204

484 differencing amplifies several noise sources already present in nighttime-light
485 data, including atmospheric variability, sensor effects, local lighting policy
486 changes, and transient illumination unrelated to durable urban change. From
487 an applied remote-sensing perspective, this makes Δ VIIRS a useful stress test
488 for observation noise and temporal instability, but not the setting in which
489 methodological differences are most clearly expressed.

490 Second, Δ WorldPop is substantially more predictable, but only under the
491 right interpretation of the split protocol. Under RandomSplit, tree-based
492 models such as Extra Trees and XGBoost reach $R^2 \approx 0.84$ with RMSE
493 around 21–22, while MLP is also competitive. Under GroupSplit, however,
494 performance drops for every model: XGBoost falls from $R^2 = 0.836$ to
495 0.527, and Extra Trees from 0.843 to 0.466. This drop is important rather
496 than incidental. It suggests that row-wise random splitting benefits from
497 grid-specific continuity across years, whereas GroupSplit better reflects the
498 intended remote-sensing deployment setting of predicting on unseen spatial
499 units.

500 Third, tail difficulty is not a small residual effect; it is the main source of
501 error inflation. Across models, RMSE_{tail} is far larger than RMSE_{all}, especially
502 under GroupSplit. This gap indicates that rare, high-magnitude changes
503 dominate practical failure even when average metrics look acceptable. For
504 urban remote sensing, these are exactly the locations where missed change is
505 most consequential. Since RandomSplit may leak location-specific structure
506 across years within the same `grid_id`, we treat **GroupSplit results as**

507 **the primary reference** for all subsequent comparisons. The benchmark
508 therefore establishes the central problem studied in the remainder of the
509 paper: under realistic spatial generalization, average fit is not sufficient if tail
510 reliability remains weak.

511 *5.2. Contrastive Learning Calibration: Main Results*

Table 8: CL calibration results (Base vs. RMSE_CL (Mix) vs. CL) reported from `results_tail_tables.xlsx`. We report R^2_{all} /RMSE_{all} for overall (body) performance and R^2_{tail} /RMSE_{tail} for tail performance. Rows are grouped by target (dWorldPop, dVIIRS) and split (GS: GroupSplit(grid_id), RS: Random(70/15/15)). Best value within each (Base, Mix, CL) triplet is marked with * at the displayed precision (ties starred).

Ridge Regression													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.288*	0.241	-0.336	47.194*	48.728	64.630	0.208	0.312*	0.253	145.595	135.709*	141.349
	RS	0.354*	0.323	-0.348	43.593*	44.623	62.960	0.366	0.447*	0.302	118.483	110.640*	124.307
dVIIRS	GS	0.025*	-0.054	-0.547	4.757*	4.945	5.991	0.030	0.035*	-0.104	13.648	13.613*	14.558
	RS	0.008*	-0.133	-0.738	4.427*	4.731	5.861	0.017*	-0.001	-0.151	12.215*	12.322	13.214
Elastic Net Regression													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.211*	0.177	-0.484	49.658*	50.727	68.117	0.052	0.239*	0.148	159.212	142.664*	150.966
	RS	0.274*	0.266	-0.472	46.197*	46.458	65.804	0.215	0.412*	0.237	131.789	114.091*	129.991
dVIIRS	GS	-0.000*	-0.006	-0.059	4.817*	4.831	4.956	-0.000*	-0.001	-0.007	13.856*	13.861	13.905
	RS	-0.000*	-0.009	-0.090	4.446*	4.464	4.641	-0.002	-0.000*	-0.004	12.332	12.319*	12.344
Support Vector Regression (SVR)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.374*	0.340	-0.007	44.234*	45.428	56.116	0.283	0.483*	0.473	138.528	117.573*	118.693
	RS	0.440	0.511*	0.260	40.597	37.910*	46.664	0.373	0.551*	0.485	117.766	99.641*	106.807
dVIIRS	GS	-0.009*	-0.326	-1.743	4.838*	5.547	7.977	0.022*	-0.090	-0.708	13.706*	14.469	18.108
	RS	-0.047*	-0.402	-1.755	4.549*	5.263	7.378	-0.014*	-0.164	-0.771	12.403*	13.287	16.392
Random Forest Regressor (RF)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.511*	0.504	0.480	39.116*	39.373	40.327	0.573	0.595	0.622*	106.848	104.055	100.549*
	RS	0.784	0.786*	0.776	25.209	25.087*	25.697	0.770	0.776*	0.769	71.417	70.385*	71.537
dVIIRS	GS	0.105*	0.048	-0.041	4.558*	4.700	4.914	0.173	0.203	0.204*	12.604	12.372	12.365*
	RS	0.066*	-0.014	-0.090	4.296*	4.476	4.641	0.144	0.157*	0.149	11.395	11.313*	11.365

Table 8: CL calibration results (continued).

Extra Trees Regressor (ETR)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.466*	0.465	0.466*	40.875	40.889	40.861*	0.513	0.514*	0.511	114.195	114.012*	114.409
	RS	0.843*	0.843*	0.843*	21.499*	21.502	21.510	0.826*	0.826*	0.826*	62.033*	62.045	62.066
dVIIRS	GS	0.093	0.094*	0.088	4.587	4.586*	4.601	0.168	0.168	0.171*	12.639	12.642	12.619*
	RS	0.014*	0.011	0.009	4.414*	4.420	4.425	0.131*	0.131*	0.131*	11.485	11.481	11.480*
Multi-layer Perceptron Regressor (MLP)													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.416*	0.360	0.025	42.739*	44.742	55.204	0.654	0.660*	0.576	96.204	95.393*	106.494
	RS	0.766*	0.756	0.435	26.239*	26.766	40.760	0.791*	0.790	0.507	68.011*	68.178	104.449
dVIIRS	GS	0.038*	-0.094	-1.301	4.725*	5.039	7.307	0.053*	0.037	-0.472	13.486*	13.596	16.813
	RS	0.010*	-0.168	-0.934	4.424*	4.805	6.182	0.020*	0.006	-0.181	12.193*	12.282	13.385
XGBoost													
Target	Split	R^2_{all}			RMSE _{all}			R^2_{tail}			RMSE _{tail}		
		Base	Mix	CL	Base	Mix	CL	Base	Mix	CL	Base	Mix	CL
dWorldPop	GS	0.527*	0.526	0.426	38.462*	38.499	42.359	0.563	0.577	0.628*	108.125	106.320	99.790*
	RS	0.820	0.823*	0.666	23.018	22.821*	31.368	0.822	0.827*	0.672	62.821	61.846*	85.215
dVIIRS	GS	0.058*	0.003	-1.018	4.675*	4.810	6.843	0.137	0.143*	-0.152	12.869	12.825*	14.872
	RS	0.050*	0.002	-1.136	4.333*	4.441	6.496	0.144*	0.140	-0.315	11.398*	11.421	14.125

512 We next ask whether tail robustness under leakage-aware spatial gen-
513 eralization is better improved by post-hoc calibration than by modifying
514 the base predictor itself. To isolate this question, we compare three vari-
515 ants while keeping the underlying regressor fixed: Base (no calibration),
516 Mix (RMSE+CL), and CL-only. Framed for applied remote sensing, this
517 comparison asks whether more reliable urban change mapping comes from
518 better-calibrated predictions or from more elaborate model structure.

519 5.2.1. How to Read Table 8

520 The table is organized by target and split (rows) and by model family and
521 calibration variant (columns). We report R^2_{all} , RMSE_{all}, R^2_{tail} , and RMSE_{tail}
522 for each configuration.

523 5.2.2. Results and Interpretation

524 Table 8 shows a clear answer: calibration helps only when it is anchored.
525 Under GroupSplit on Δ WorldPop, the mixed objective delivers the most
526 reliable tail improvements while largely preserving overall fit. The pattern
527 appears across different model families. For SVR, R^2_{tail} rises from 0.283 to

0.483 and tail RMSE drops from 138.528 to 117.573, despite only a modest decline in overall R^2 from 0.374 to 0.340. Random Forest improves from 0.573 to 0.595 in R^2_{tail} and from 106.848 to 104.055 in $\text{RMSE}_{\text{tail}}$. XGBoost, our strongest baseline, also benefits, with R^2_{tail} increasing from 0.563 to 0.577 and $\text{RMSE}_{\text{tail}}$ decreasing from 108.125 to 106.320.

The fact that gains concentrate in tail metrics rather than global metrics is itself informative. The calibrator is not acting as a generic rescaling layer; instead, it mainly corrects the underestimation of rare high-magnitude outcomes. This is especially valuable in long-tail regression, where strong base models often fit the body reasonably well but still regress extreme cases toward the center. For remote-sensing proxy products, that behavior translates into muted maps of rapid local change. A lightweight affine layer has enough flexibility to adjust this bias without relearning the entire predictor.

The magnitude of improvement also depends on the base model. Underfit or more biased models such as Ridge, Elastic Net, and SVR tend to benefit more because their tail predictions are systematically shrunk toward the mean, leaving more room for correction. By contrast, Extra Trees and some Random Forest configurations show smaller gains because their baseline fit is already strong and their prediction scale is closer to the target scale. In these cases, the calibrator still helps selectively, but the ceiling for post-hoc improvement is lower.

By comparison, CL-only is often unstable. Table 9 shows that CL-only can improve tail metrics, but it does so by paying too much in overall fit: under GroupSplit on $\Delta\text{WorldPop}$, average tail R^2 increases by +40.17%, yet average overall R^2 drops by -109.59% . This instability is consistent with the geometry of the objective. Pure contrastive optimization encourages separation between predictions and incorrect targets, but without an anchoring term it can overstretch the body of the distribution, causing large distortions in pointwise accuracy. The RMSE term prevents this drift and makes the calibration practically usable.

RandomSplit exhibits the same qualitative ordering, with Mix generally outperforming Base and CL-only remaining erratic. However, because RandomSplit may preserve grid-specific continuity across train and test sets, we treat the GroupSplit results as the main evidence. Under the leakage-aware split that best matches deployment, the practical conclusion is consistent: anchored calibration improves tail reliability, whereas unanchored calibration does not.

Table 9: Average improvements of calibration variants over Base on **dWorldPop** under **GS**, aggregated across **seven models**. For R^2 , gain is Var – Base; for RMSE, gain is Base – Var (so larger is better). Percentage gains are computed *per model* and then averaged.

Metric	Base (avg)	Gain (avg)	Gain (% avg)
Mix (RMSE_CL) vs. Base			
R^2_{all}	0.399	−0.0257	−8.11%
RMSE _{all}	43.183	−0.873	−1.94%
R^2_{tail}	0.407	+0.0763	+69.68%
RMSE _{tail}	124.101	+7.569	+5.37%
CL-only vs. Base			
R^2_{all}	0.399	−0.3176	−109.59%
RMSE _{all}	43.183	−9.334	−20.48%
R^2_{tail}	0.407	+0.0521	+40.17%
RMSE _{tail}	124.101	+5.208	+3.59%

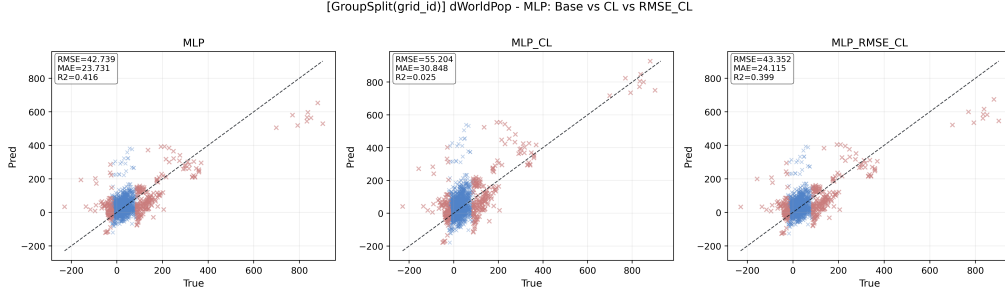


Figure 3: Visualization of contrastive learning adjustment: a representative example showing how contrastive learning improves predictions for tail samples. In the middle graph (MLP_CL), an upper-right cluster of tail sample predictions improves substantially, while body predictions are stretched by the adjusting layer, leading to a decrease in overall performance.

5.3. Spike Diagnostics and Error Analysis

We further examine why the calibrator helps. Figure 3 shows that calibration mainly changes the geometry of tail predictions rather than uniformly shifting the whole cloud. Tail samples in the upper-right region move closer to the diagonal, indicating that the base model had been systematically conservative on large positive changes. This is a common failure mode in

571 long-tail regression: when rare shocks are underrepresented, the fitted model
572 prefers safer predictions near the center of the distribution.

573 At the same time, the figure also reveals the main risk of aggressive con-
574 trastive adjustment: some body predictions are pulled away from the diagonal.
575 This is why anchoring matters. With the mixed objective, the calibrator
576 can correct tail shrinkage while the RMSE term discourages unnecessary
577 movement of already well-predicted body samples. In other words, the affine
578 layer works not because it is expressive, but because it provides a controlled
579 correction to a recurrent bias in the base predictors.

580 **Summary:** the calibration gains are consistent with a simple mechanism:
581 rare extremes are systematically underfit, and a lightly constrained post-
582 hoc layer can correct that bias as long as it remains anchored to pointwise
583 accuracy.

584 5.4. Ablations (CL Calibration)

585 We evaluate how the mix weight changes the calibration behavior. Empir-
586 ically, a setting around 5% RMSE and 95% CL provides the best trade-off:
587 larger RMSE weights reduce the tail-focused effect, while removing RMSE
588 entirely makes the calibrator unstable. This supports the view that the
589 method succeeds not by replacing standard regression objectives, but by
590 supplementing them with a tail-sensitive correction.

591 Some models, especially Extra Trees and stronger Random Forest variants,
592 show only modest gains. This is not a contradiction. When a baseline already
593 captures much of the target scale and nonlinearity, the remaining errors are
594 less systematic and therefore less correctable by a one-dimensional affine map.
595 The calibrator is most valuable when there is a persistent tail bias to correct.

596 5.5. MoE: Negative Results

597 We next test the competing hypothesis that tail robustness should be
598 addressed by routing-based decomposition. In principle, a Mixture-of-Experts
599 should help if tail and body samples correspond to separable regimes with
600 distinct predictive mechanisms. In our setting, however, the opposite occurs.

601 Table 10 shows that the two-expert MoE under GroupSplit on Δ WorldPop
602 degrades performance relative to a single XGBoost baseline across all metrics:

603 This result is informative rather than incidental. The MoE introduces a
604 harder subproblem than the original regression task: it must first infer whether
605 a sample belongs to the tail regime before either expert can help. Under
606 spatial shift, that routing decision is weakly identifiable from the available

Table 10: MoE performance degradation on Δ WorldPop, GroupSplit(grid_id). For R^2 : gain = $(\text{MoE} - \text{Base})/\text{Base} \times 100\%$ (higher is better). For RMSE: gain = $(\text{Base} - \text{MoE})/\text{Base} \times 100\%$ (higher is better). Negative values indicate degradation.

Metric	Base	MoE	Gain (%)
Base (XGB) vs. MoE (2-expert)			
R^2_{all}	0.527	0.420	−20.30%
RMSE_{all}	38.462	46.474	−20.83%
R^2_{tail}	0.563	0.321	−42.98%
$\text{RMSE}_{\text{tail}}$	108.125	137.703	−27.35%

features. Figure 4 shows broad ambiguity near the decision boundary, where many samples receive intermediate routing probabilities rather than confident assignments. For a remote-sensing application, this means the model struggles to decide when an observation should be interpreted as an exceptional local change rather than ordinary variability.

Three mechanisms likely drive the degradation. First, routing errors are expensive because the tail regime is rare and mistakes on extreme samples dominate $\text{RMSE}_{\text{tail}}$. Second, splitting the data between experts reduces effective sample size, which is especially harmful for a scarce tail regime. Third, the signed-log transformation used for the tail expert may stabilize training locally, but its inverse magnifies small prediction errors when mapped back to the original scale. As a result, the added complexity does not improve specialization; it amplifies instability.

6. Discussion

We evaluated machine learning models for predicting population and nighttime luminosity changes using CivitasGrid-TLP. From an applied remote-sensing perspective, the central result is that reliability under spatial transfer matters more than optimistic average performance under permissive splits. RMSE-anchored contrastive calibration substantially improves tail performance without sacrificing overall accuracy, whereas the two-expert MoE unexpectedly underperformed. We also found that Δ VIIRS is substantially more challenging to predict than Δ WorldPop, likely reflecting increased noise from differencing. We discuss these findings in detail below.

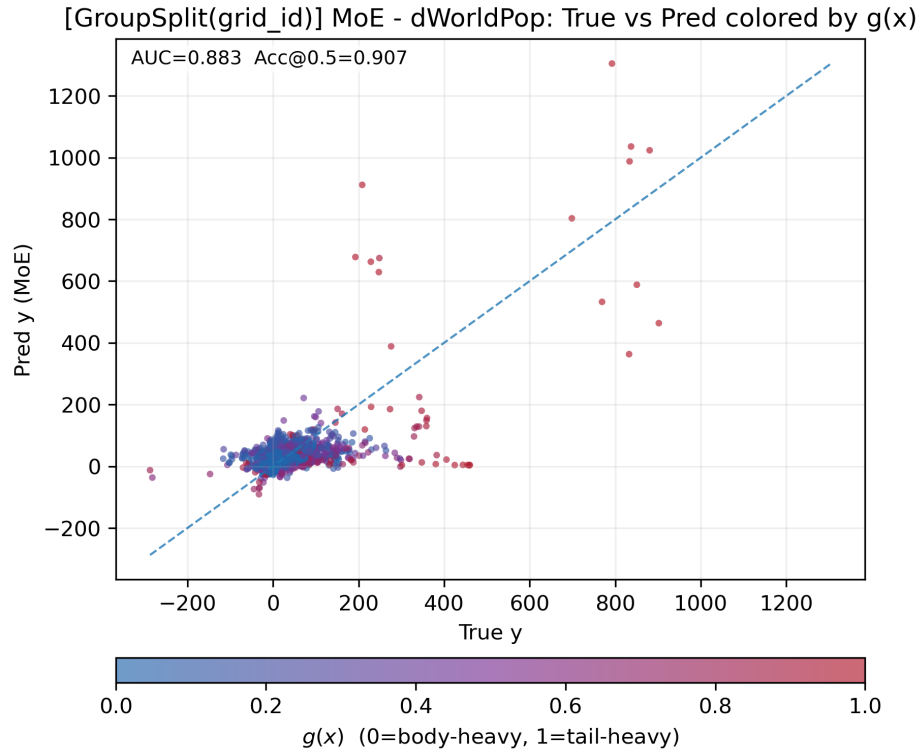


Figure 4: Gating network scatter plots showing the distribution of soft routing probabilities. The gating network attempts to differentiate body and tail samples but exhibits weak separability near the decision boundary, leading to ambiguous expert assignments.

6.1. Cross-Target Performance Differences

Δ WorldPop predictions consistently outperform Δ VIIRS across all models and metrics. Multiple factors explain this difference. First, nighttime light data generally contain more uncertainty and noise from atmospheric conditions, sensor degradation, and local lighting policies, whereas population data exhibit more stable patterns. Second, year-to-year differencing amplifies noise, which is particularly problematic for VIIRS because it is already noisy. Third, our transportation morphology features may be more informative for population changes than for light intensity changes, suggesting potential benefit from target-specific feature engineering. In other words, the two targets occupy different positions on the remote-sensing signal-to-noise spectrum, and they should not be expected to respond equally well to the same modeling strategy.

6.2. RMSE+CL Calibration: Mechanisms and Evidence

The RMSE+CL mixed objective delivers consistent tail improvements. We attribute this to three mechanisms that are especially relevant when remote-sensing proxy products are used for change-focused mapping:

6.2.1. Anchoring

The RMSE component prevents the contrastive loss from over-emphasizing tail samples at the expense of body accuracy. By normalizing both terms and weighting them at 5% RMSE and 95% CL, the calibrator avoids the instability observed with CL-only (which frequently collapsed overall R^2 to negative values in Table 8).

6.2.2. Dual Optimization

RMSE minimization preserves pointwise accuracy while contrastive learning encourages separation between predictions and incorrect targets. This naturally pushes tail predictions (far from the mean) along the regression line, improving reliability without harming body predictions.

6.2.3. Empirical Evidence

Under GroupSplit on Δ WorldPop, RMSE+CL achieves a +69.68% average tail R^2 gain while changing overall RMSE by only -1.94% (Table 9). This asymmetry shows effective rebalancing toward rare events without degradation. In contrast, CL-only sacrifices -109.59% overall R^2 for only $+40.17\%$ tail R^2 , indicating that pure contrastive learning is too aggressive.

664 6.3. MoE Failure Analysis

665 The two-expert MoE underperforms across both overall and tail metrics,
666 highlighting challenges for routing-based approaches in long-tail settings.

667 First, predicting tail membership from available features is intrinsically
668 difficult: the gating network struggles near quantile thresholds, where samples
669 are inherently ambiguous. Small feature changes can flip expert assignments,
670 amplifying variance and creating boundary artifacts. Second, misrouting is
671 costly: routing a tail sample to the body expert (or vice versa) produces much
672 larger errors than single-model baselines, which is especially problematic when
673 rare extremes dominate tail metrics. Third, while the signed-log transforma-
674 tion stabilizes tail expert training, the inverse transformation magnifies small
675 errors into large absolute values for extreme predictions.

676 These findings indicate that simple binary MoE with soft/hard gating is
677 ill-suited for this task. More robust approaches—uncertainty-aware gating,
678 continuous mixtures, or threshold-aware routing—may better handle the
679 ambiguity and high stakes of long-tail routing in Earth-observation settings.

680 6.4. Practical Recommendations

681 For long-tailed targets where tail reliability is critical, we recommend
682 adding a post-hoc RMSE+CL calibration layer to strong baselines such as
683 XGBoost or Random Forest. This lightweight, model-agnostic approach
684 effectively enhances tail predictions while preserving overall performance,
685 making it practical for remote-sensing practitioners who need geographically
686 transferable urban proxy maps rather than only strong in-sample scores.

687 6.5. Limitations and Future Work

688 We acknowledge several limitations and offer practical directions for future
689 work:

690 (1) *VIIRS* prediction remains unsatisfactory. Future work should explore
691 smoothed targets (e.g., three-year trends) to reduce differencing noise, suppress
692 transient illumination effects, and improve signal.

693 (2) Our model predicts point estimates without uncertainty. Quantile
694 regression or Bayesian approaches could enable uncertainty-aware prediction,
695 improving decision support for remote-sensing monitoring workflows.

696 (3) MoE’s failure may stem from weak gating networks. Future work
697 should explore uncertainty-aware or threshold-informed routing mechanisms.

698 (4) Spatial bias from OSM coverage differences could be mitigated through
699 coverage-weighted training or post-hoc bias correction.

700 7. Conclusion

701 Urban proxy dynamics are strongly heavy-tailed: most grid-year changes
 702 remain near zero, while rare shocks dominate error and risk. To study
 703 this regime reproducibly, we introduced CivitasGrid-TLP, a grid-level panel
 704 dataset (2015–2024) that integrates OSM transportation morphology with
 705 proxy socioeconomic signals (WorldPop and VIIRS). Framed as an applied
 706 remote-sensing task, the dataset supports evaluation of how well models can
 707 transfer urban proxy mapping to unseen grids rather than simply interpolate
 708 within previously observed locations. We evaluated models under both random
 709 splitting (potentially optimistic) and a leakage-resistant GroupSplit(grid_id)
 710 protocol.

711 Key findings across baselines: Δ WorldPop is substantially more pre-
 712 dictable than Δ VIIRS, whose year-to-year differencing amplifies measurement
 713 noise, yielding near-zero R^2 for most methods. For long-tail robustness, our
 714 primary contribution is demonstrating that a lightweight, model-agnostic
 715 post-hoc affine calibrator trained with an *RMSE-anchored contrastive* objec-
 716 tive consistently improves tail performance while preserving overall accuracy.
 717 Pure contrastive loss optimization is often unstable and severely degrades
 718 global fit. We also report a negative result: a two-expert MoE (body expert
 719 in original space, tail expert in signed-log space) underperforms a single XG-
 720 Boost baseline on both overall and tail metrics, likely due to weak tail-routing
 721 identifiability and amplified transformation errors.

722 **Takeaway:** reliable tail gains require anchoring. For practitioners needing
 723 improved tail accuracy in remote-sensing proxy mapping, RMSE-anchored
 724 contrastive calibration is a practical, lightweight add-on to strong tabular
 725 regressors. Naive tail/body MoE routing can collapse under feature–label
 726 ambiguity near quantile boundaries.

727 Future work should address VIIRS noise (via temporal smoothing or trend
 728 targets), incorporate uncertainty quantification (quantile or distributional
 729 regression), and explore uncertainty-aware or threshold-informed routing
 730 mechanisms to improve mixture-model stability in long-tail remote-sensing
 731 settings.

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737 **Declaration of competing interest**

738 The author declares no competing interests.

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741 **Data availability**

742 CivitasGrid-TLP is publicly available on Zenodo (DOI 10.5281/zen-
743 odo.18144416) [10].

744 **Code availability**

745 Code is available from the author upon reasonable request.

746 **Declaration of generative AI and AI-assisted technologies in the** 747 **manuscript preparation process**

748 During the preparation of this work the author used ChatGPT in order
749 to improve language polishing and debug LaTeX issues. After using this
750 tool, the author reviewed and edited the content as needed and takes full
751 responsibility for the content of the published article.

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787 **Appendix A. Appendix**

Table A.11: Hyperparameters used in Contrastive Learning experiments

Component	Hyperparameter	Value
Global	Random seed	SEED=42
	Splitting ratios	Train/Val/Test = 70% / 15% / 15%
	Group split	GroupShuffleSplit; two-stage: hold out test groups, then split remaining groups into train/val
Tail (for reporting)	Tail quantiles	q05=0.05, q95=0.95 (computed on train split only)
	Tail mask	$y \leq q_{0.05}$ or $y \geq q_{0.95}$
Baselines (sklearn)	Ridge	$\alpha = 1.0$
	ElasticNet	$\alpha = 1.0$, l1_ratio=0.5, max_iter=5000, random_state=42
	SVR (RBF)	$C = 10.0$, $\epsilon = 0.1$, kernel=rbf
	RandomForestRegressor	n_estimators=150, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1
	ExtraTreesRegressor	n_estimators=300, max_depth=None, min_samples_split=2, min_samples_leaf=1, random_state=42, n_jobs=-1
	MLPRegressor	hidden_layer_sizes=(128,64), activation=relu, early_stopping=True, validation_fraction=0.1, n_iter_no_change=15
CL calibrator (affine)	Calibrator form	$\tilde{y} = a\hat{y} + b$ (trainable a,b)
	Mode	cl (contrastive-only) or rmse_cl (mixed)
	CL batch size	CL_BATCH_SIZE=128
	Epochs	CL_EPOCHS=40
	Learning rate	CL_LR=0.05
	Temperature	$T = \max(\text{Std}(y_{\text{train}}) \cdot \text{CL_TEMP_FACTOR}, 10^{-3})$, CL_TEMP_FACTOR=1.0
	Numerical eps (loss)	CL_EPS=1e-12
	Mixed-loss weights	rmse_weight=0.05, cl_weight=0.95
	Normalization (rmse0)	rmse0 = RMSE on train with initial (a,b)=(1,0)
	Normalization (cl0)	Subsample estimate: CL0_MAX_N=3000, CL0_BATCH=512
	Adam update eps	eps=1e-8